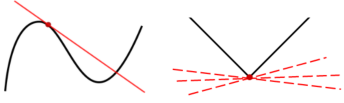
Optimization in Machine Learning

Mathematical Concepts
Differentiation and Derivatives



Learning goals

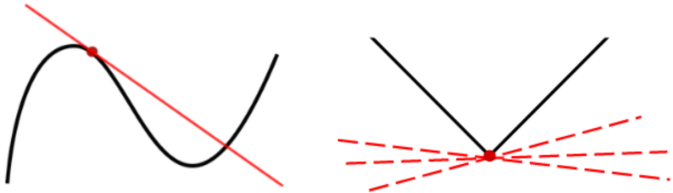
- Definition of smoothness
- Uni- & multivariate differentiation
- Gradient, partial derivatives
- Jacobian matrix
- Hessian matrix
- Lipschitz continuity

UNIVARIATE DIFFERENTIABILITY

Definition: A function $f : \mathcal{S} \subseteq \mathbb{R} \to \mathbb{R}$ is said to be **differentiable** for each inner point $x \in \mathcal{S}$ if the following limit exists:

$$f'(x) := \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

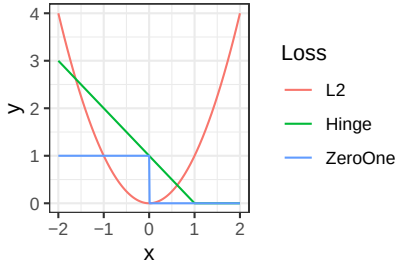
Intuitively: f can be approximated locally by a lin. fun. with slope $m = f'(x)$.



Left: Function is differentiable everywhere. **Right:** Not differentiable at the red point.

SMOOTH VS. NON-SMOOTH

- **Smoothness** of a function $f : \mathcal{S} \to \mathbb{R}$ is measured by the number of its continuous derivatives
- $\mathcal{C}^k$ is class of k-times continuously differentiable functions ($f \in \mathcal{C}^k$ means $f^{(k)}$ exists and is continuous)
- In this lecture, we call f "smooth", if at least $f \in \mathcal{C}^1$



f_1 is smooth, f_2 is continuous but not differentiable, and f_3 is non-continuous.

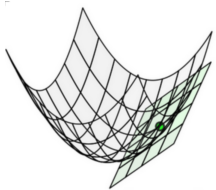
MULTIVARIATE DIFFERENTIABILITY

Definition: For a function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ of d variables $x_1, \ldots, x_d$, **partial derivatives** are defined as

$$\frac{\partial f}{\partial x_1} = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \ldots, x_d) - f(\boldsymbol{x})}{h}$$

$$\vdots$$

$$\frac{\partial f}{\partial x_d} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{d-1}, x_d + h) - f(\boldsymbol{x})}{h}$$



Geometrically: Similarly to the 1D case, the vector of partial derivatives can be used to determine a tangent hyperplane. ▸ Click for source

GRADIENT

- Specifically, the vector of partial derivatives is called the **gradient**:

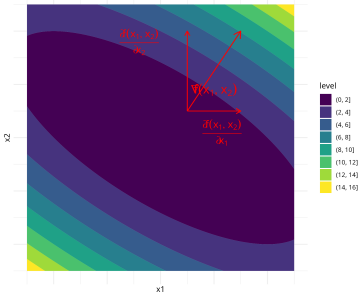
$$\nabla_{\boldsymbol{x}} f \text{ or } \nabla f := \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_d} \right)$$

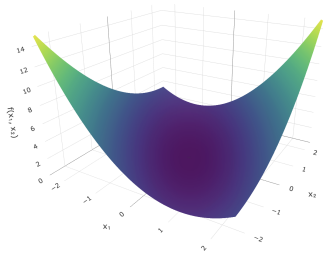
  note that this is a row vector!

- This gradient of f can be used to linearly approximate f:

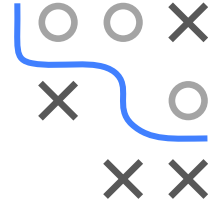
$$f(\boldsymbol{x} + \mathbf{h}) = f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})\mathbf{h} + o(\mathbf{h})$$

- **Example:** $f(\boldsymbol{x}) = x_1^2/2 + x_1 x_2 + x_2^2 \Rightarrow \nabla f(\boldsymbol{x}) = (x_1 + x_2, x_1 + 2x_2)$





DIRECTIONAL DERIVATIVE



The **directional derivative** tells how fast $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ is changing w.r.t. an arbitrary direction $\boldsymbol{v}$:

$$D_{\boldsymbol{v}} f(\boldsymbol{x}) := \lim_{h \to 0} \frac{f(\boldsymbol{x} + h\boldsymbol{v}) - f(\boldsymbol{x})}{h} = \nabla f(\boldsymbol{x}) \boldsymbol{v}$$

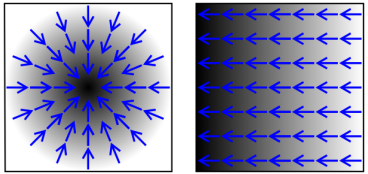
Example: The directional derivative for $\boldsymbol{v} = (1, 1)$ is:

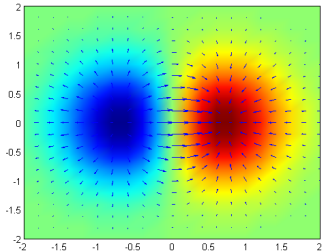
$$\begin{aligned} D_{\boldsymbol{v}} f(\boldsymbol{x}) &= \nabla f(\boldsymbol{x}) \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{\partial f}{\partial x_1} + \frac{\partial f}{\partial x_2} \end{aligned}$$

NB: Some people require that $||\boldsymbol{v}|| = 1$. Then, we can identify $D_{\boldsymbol{v}} f(\boldsymbol{x})$ with the instantaneous rate of change in direction $\boldsymbol{v}$, i.e. $\lim_{h \to 0} \frac{f(\boldsymbol{x}+h\boldsymbol{v}) - f(\boldsymbol{x})}{h}$ – and in our example we would have to divide by $\sqrt{2}$.

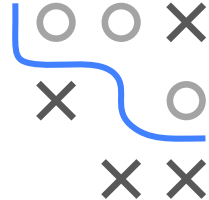
IMPORTANT PROPERTIES OF THE GRADIENT

- **Orthogonal** to level curves/surfaces of a function
- Points in direction of **greatest increase** of f





IMPORTANT PROPERTIES OF THE GRADIENT



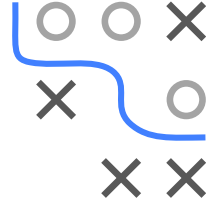
- To prove 1, consider that a *level curve* of a function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ is any parametrized, differentiable curve (see *implicit function theorem*) $x(t)$, $t \in \mathcal{T} \subseteq \mathbb{R}_{\geq 0}$ so that $f(x(t)) = c$ for some constant c.
- **Proof of 1.**: Consider a point $x_0 \in \mathbb{R}^d$ so that $f(x_0) = c$ and $\exists t_0 \in \mathcal{T} : \; x(t_0) = x_0$. To prove orthogonality, we want to show $\langle \nabla_{\boldsymbol{x}} f(x_0), \frac{dx}{dt}(t_0) \rangle = 0$.
- Note that

$$\left\langle \nabla_{\boldsymbol{x}} f(x_0), \frac{dx}{dt}(t_0) \right\rangle = \nabla_{\boldsymbol{x}} f(x(t_0)) \frac{dx}{dt}(t_0) = \frac{d\,(f \circ x)}{dt}(t_0)$$

  By the chain rule, see slide-set “Matrix Calculus”.

- Since $f \circ x$ is a constant function in t, the derivative will always equal zero and so $\frac{d\,(f \circ x)}{dt}(t_0) = 0$, proving the statement.

IMPORTANT PROPERTIES OF THE GRADIENT

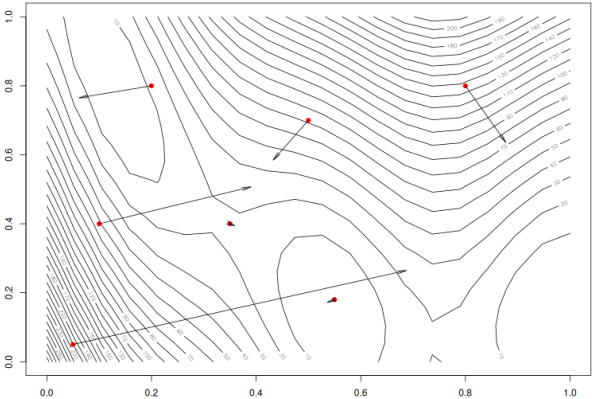


- **Proof of 2.**: Let $\boldsymbol{v}$ be a vector with $\|\boldsymbol{v}\| = 1$ and θ the angle between $\boldsymbol{v}$ and $\nabla f(\boldsymbol{x})$.

$$D_{\boldsymbol{v}} f(\boldsymbol{x}) = \nabla f(\boldsymbol{x})\boldsymbol{v} = \|\nabla f(\boldsymbol{x})\| \, \|\boldsymbol{v}\| \cos(\theta) = \|\nabla f(\boldsymbol{x})\| \cos(\theta)$$

  by the cosine formula for dot products and $\|\boldsymbol{v}\| = 1$.
- $\cos(\theta)$ is maximal if $\theta = 0$, hence if $\boldsymbol{v}$ and $\nabla f(\boldsymbol{x})$ point in the same direction.
- (Alternative proof: Apply Cauchy-Schwarz to $\nabla f(\boldsymbol{x})\boldsymbol{v}$ and look for equality.)
- Analogous: Negative gradient $-\nabla f(\boldsymbol{x})$ points in direction of greatest *de*crease

IMPORTANT PROPERTIES OF THE GRADIENT



Length of arrows is norm of their gradient.

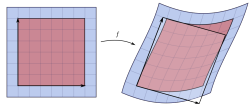
JACOBIAN MATRIX

For vector-valued function $f : \mathcal{S} \subseteq \mathbb{R}^d \longrightarrow \mathbb{R}^m,\ \boldsymbol{x} \mapsto (f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x}))^\top$, $f_j : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$, the **Jacobian** matrix $J_f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^{m \times d}$ generalizes gradient by placing all ∇f_j in its rows:

$$J_f(\boldsymbol{x}) \text{ or } \nabla f(\boldsymbol{x}) = \begin{pmatrix} \nabla f_1(\boldsymbol{x}) \\ \vdots \\ \nabla f_m(\boldsymbol{x}) \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\boldsymbol{x})}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(\boldsymbol{x})}{\partial x_1} & \cdots & \frac{\partial f_m(\boldsymbol{x})}{\partial x_d} \end{pmatrix}$$

We will mainly use the ∇f notation.

- Jacobian gives best linear approximation of distorted volumes



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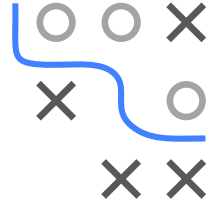
JACOBIAN DETERMINANT



- Let $f \in \mathcal{C}^1$ and $\boldsymbol{x}_0 \in \mathcal{S} \subseteq \mathbb{R}^d$.
- **Inverse function theorem:** Let $\boldsymbol{y}_0 = f(\boldsymbol{x}_0)$. If $\det(J_f(\boldsymbol{x}_0)) \neq 0$, then
  1. f is invertible in a neighborhood of $\boldsymbol{x}_0$,
  2. $f^{-1} \in \mathcal{C}^1$ with $J_{f^{-1}}(\boldsymbol{y}_0) = J_f(\boldsymbol{x}_0)^{-1}$.
- $|\det(J_f(\boldsymbol{x}_0))|$: factor by which f expands/shrinks volumes near $\boldsymbol{x}_0$
- If $\det(J_f(\boldsymbol{x}_0)) > 0$, f preserves orientation near $\boldsymbol{x}_0$
- If $\det(J_f(\boldsymbol{x}_0)) < 0$, f reverses orientation near $\boldsymbol{x}_0$

HESSIAN MATRIX

For real-valued function $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$, the **Hessian** matrix $\nabla^2 : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^{d \times d}$ contains all its second derivatives (if they exist):

$$\nabla^2 f(\boldsymbol{x}) = \left(\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} \right)_{i,j=1,\ldots,d}$$

Note: Hessian of f is Jacobian of ∇f. Also, the Hessian is often denoted by $H(\boldsymbol{x}) \hat{=} \nabla^2 f(\boldsymbol{x})$. **Example**: Let $f(\boldsymbol{x}) = \sin(x_1) \cdot \cos(2x_2)$.

Then:

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} -\cos(2x_2) \cdot \sin(x_1) & -2\cos(x_1) \cdot \sin(2x_2) \\ -2\cos(x_1) \cdot \sin(2x_2) & -4\cos(2x_2) \cdot \sin(x_1) \end{pmatrix}$$

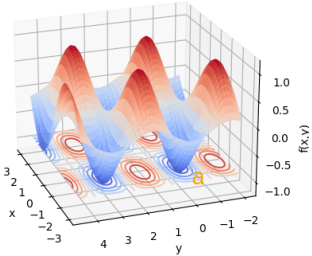
- If $f \in \mathcal{C}^2$, then $\nabla^2 f$ is symmetric
- Many local properties (geometry, convexity, critical points) are encoded by the Hessian and its spectrum ($\to$ later)

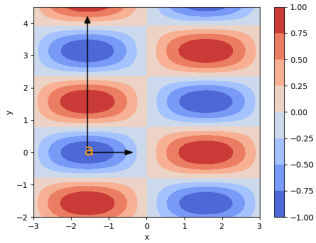
LOCAL CURVATURE BY HESSIAN

Eigenvector corresponding to largest (resp. smallest) **eigenvalue** of Hessian points in direction of largest (resp. smallest) **curvature**
Example (previous slide)**:** For $\boldsymbol{a} = (-\pi/2, 0)^T$, we have

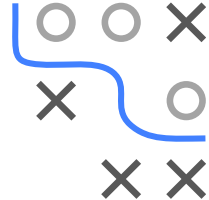
$$\nabla^2 f(\boldsymbol{a}) = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

and thus $\lambda_1 = 4, \lambda_2 = 1$, $\boldsymbol{v}_1 = (0, 1)^T$, and $\boldsymbol{v}_2 = (1, 0)^T$.





LIPSCHITZ CONTINUITY



Function $h : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}^m$ is **Lipschitz continuous** if slopes are bounded:

$$\|h(\boldsymbol{x}) - h(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\| \quad \text{for each } \boldsymbol{x}, \boldsymbol{y} \in \mathcal{S} \subseteq \mathbb{R}^d \text{ and some } L > 0$$

- **Examples** ($d = m = 1$)**:** $\sin(x)$, $|x|$
- **Not** examples: $1/x$ (but *locally* Lipschitz continuous), $\sqrt{x}$
- If $m = d$ and h **differentiable**:

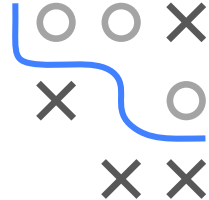
$$h \text{ Lipschitz continuous with constant } L \iff J_h \preccurlyeq L \cdot \mathbf{I}_d$$

Note: $\mathbf{A} \preccurlyeq \mathbf{B} :\iff \mathbf{B} - \mathbf{A}$ is positive semidefinite, i.e., $\mathbf{v}^T(\mathbf{B} - \mathbf{A})\mathbf{v} \geq 0 \;\; \forall \mathbf{v} \neq 0$.
Proof of "$\Rightarrow$" for $d = m = 1$**:**

$$h'(x) = \lim_{\epsilon \to 0} \frac{h(x+\epsilon) - h(x)}{\epsilon} \leq \lim_{\epsilon \to 0} \left| \frac{h(x+\epsilon) - h(x)}{\epsilon} \right| \leq \lim_{\epsilon \to 0} L = L$$

[**Proof** of "$\Leftarrow$" by mean value theorem: Show that $\lambda_{\max}(J_h) \leq L$.]

LIPSCHITZ GRADIENTS



- Let $f \in \mathcal{C}^2$. Since $\nabla^2 f$ is Jacobian of $h = \nabla f$ $(m = d)$:

$$\nabla f \text{ Lipschitz continuous with constant } L \iff \nabla^2 f \preccurlyeq L \cdot \mathbf{I}_d$$

- Equivalently, eigenvalues of $\nabla^2 f$ are bounded by L
- **Interpretation:** Curvature in any direction is bounded by L
- Lipschitz gradients occur frequently in machine learning $\implies$ Fairly **weak assumption**
- Important for analysis of **gradient descent** optimization $\implies$ Descent lemma (later)